FULL-AUTO AIR CONDITIONER SYSTEM [FULL-AUTO AIR CONDITIONER]

id0740a1004000

Outline

- The climate control unit performs the following controls based on the input signals from the each dial/switch and each sensor.

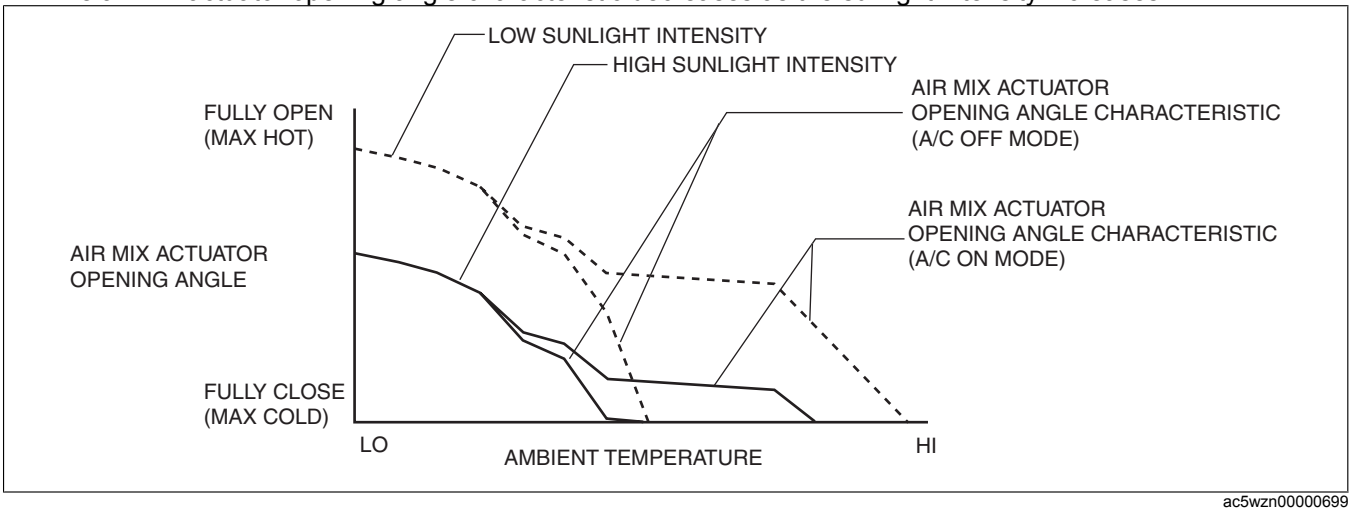
Control item	Control description
Airflow temperature control	• Airflow temperature automatic control — MAX HOT and MAX COLD correction — Engine coolant temperature correction
Airflow volume control	• Airflow volume automatic control • Airflow volume manual control — Engine coolant temperature correction (warm-up correction) — Mild start correction — MAX HOT and MAX COLD correction — Start-up window fogging prevention correction — Defroster correction
Airflow mode control	• Airflow mode automatic control • Airflow mode manual control — Engine coolant temperature correction (warm-up correction)
Air intake control	• Air intake manual control — Air intake actuator motor protection function — RECIRCULATE mode control under severe heat — FRESH control in defroster mode
A/C compressor control	• A/C compressor automatic control • A/C compressor manual control — Defroster correction • A/C cut-off control

Supplementary function
Sensor signal delay function
Air conditioner i-stop control (With i-stop)
Fail-safe
On-board diagnostic function (See ON-BOARD DIAGNOSTIC.)

Function

Airflow temperature control

- The airflow temperature control operates the air mix actuator in conjunction with the set temperature and vehicle conditions, and switches the air mix door position.
- The climate control unit calculates the opening angle characteristic of the air mix actuator based on the set temperature, solar radiation amount, and the airflow mode.
- The air mix actuator opening angle characteristic decreases as the sunlight intensity increases.



ac5wzn00000699

- The airflow temperature is automatically controlled continuously.
- The following are the airflow temperature control corrections.
  - — MAX HOT and MAX COLD correction
  - — Engine coolant temperature correction

MAX HOT and MAX COLD correction

- When the temperature is set to **29/84**, the air mix actuator opening angle is fixed at fully open, and when set to **15/59**, it is fixed at fully closed (European (L.H.D. U.K.) specs.).
- When the temperature is set to **32/90**, the air mix actuator opening angle is fixed at fully open, and when set to **18/64**, it is fixed at fully closed (except European (L.H.D. U.K.) specs.).

Engine coolant temperature correction

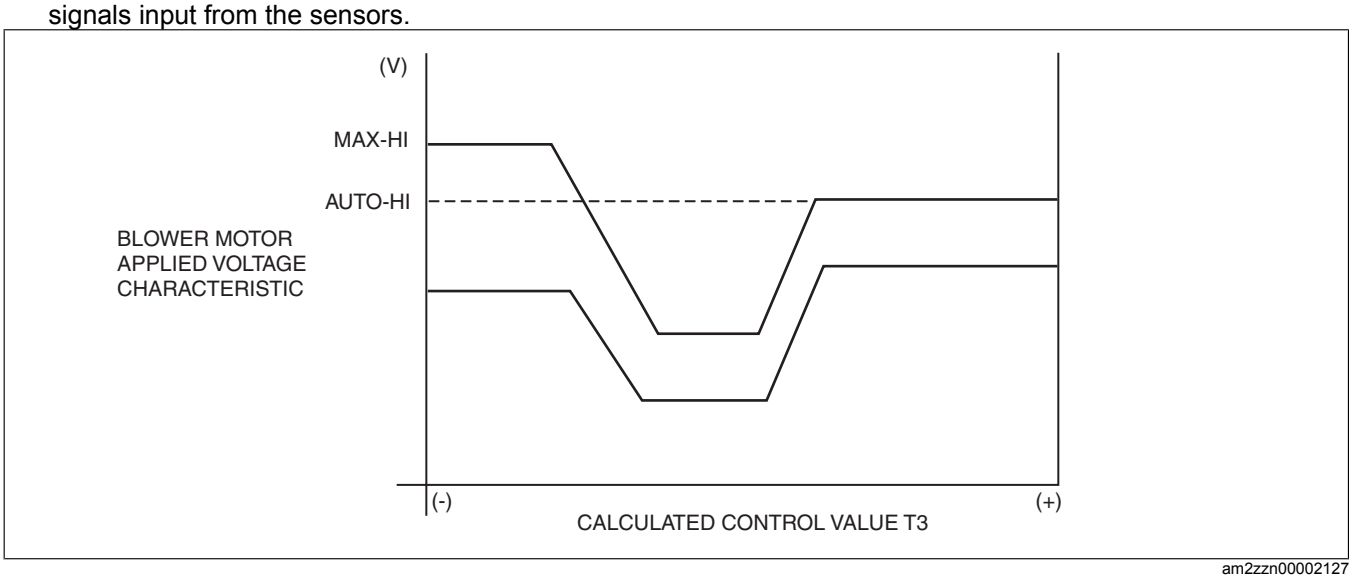
- The engine coolant temperature correction (warm-up correction) sets the opening angle of the air mix actuator more to the HOT side than during normal control when the engine coolant temperature is lower than the specified value to prevent discomfort caused by a high volume of cold air blown from the vents after the engine is started in winter.
- The engine coolant temperature correction is not performed when the ambient temperature is **15 °C {59 °F} or more**.

Airflow volume control

- The airflow volume control changes the operation of the blower motor according to the operations of the airflow volume control dial or temperature setting dial and the vehicle environment to change the airflow volume.
- The airflow volume control has automatic and manual controls.
- The airflow volume automatic control performs the following corrections:
  - — Engine coolant temperature correction (warm-up correction)
  - — Mild start correction
  - — MAX HOT and MAX COLD correction
  - — Start-up window fogging prevention correction
  - — Defroster correction

Airflow volume automatic control

- The climate control unit calculates the blower motor applied voltage characteristic based on the set temperature and ambient temperature.
- The climate control unit compares the blower motor applied voltage characteristic with the target temperature (calculated control value T3) and then determines the blower motor applied voltage (AUTO voltage).
- The calculated control value T3 is the target cabin temperature as set by the climate control unit based on the difference between the set temperatures and temperatures input from the sensors.
- The calculated control value T3 is constantly calculated according to the changes in the set temperature and the signals input from the sensors.



am2zzn00002127

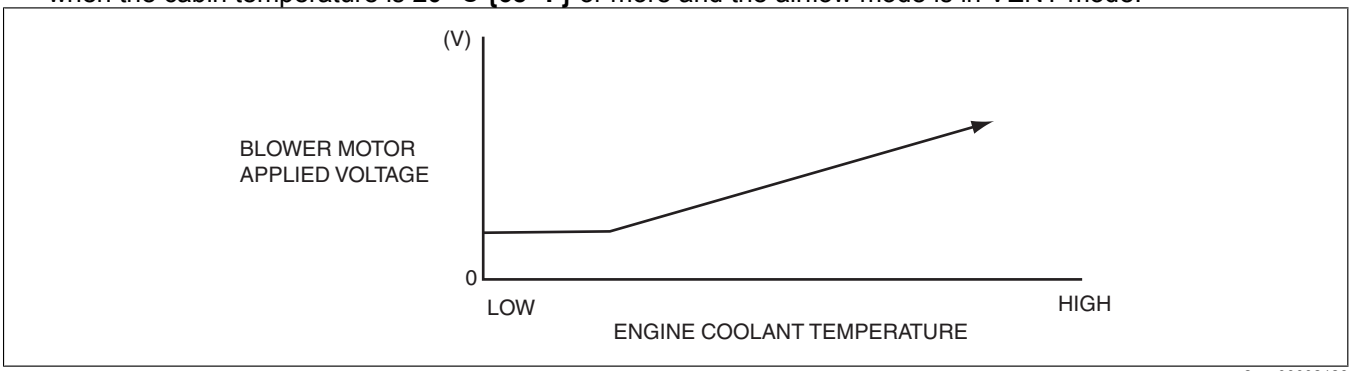
Airflow volume manual control

- The climate control unit changes the blower motor applied voltage in seven steps according to the operation of the airflow volume control dial.

Airflow volume setting	Blower motor applied voltage (reference)
1st	3.8 V
2nd	5.1V
3rd	6.7 V
4th	8.4 V
5th	10.1 V
6th	12.0 V
7th	14.6 V

Engine coolant temperature correction (warm-up correction)

- The engine coolant temperature correction (warm-up correction) controls the blower motor applied voltage in conjunction with the increase in the engine coolant temperature to prevent a high volume of cold air from flowing out of the air vents after starting the engine in the winter.
- The engine coolant temperature correction (warm-up correction) is not performed during defroster correction, or when the cabin temperature is **20 °C {68 °F}** or more and the airflow mode is in VENT mode.



am2zzn00002128

Mild start correction

- The mild start correction prevents a high volume of warm air from blowing out of the air vents when the blower motor is started.

MAX HOT and MAX COLD correction

- The MAX HOT and MAX COLD correction fixes the blower motor applied voltage as indicated in the table when the temperature is set to MAX HOT or MAX COLD.
- The MAX HOT correction is not performed during engine coolant temperature correction (warm-up correction).

European (L.H.D. U.K.) specs.

Correction	Set temperature	Blower motor applied voltage
MAX HOT correction	29/84	12.1 (V): AUTO-HI
MAX COLD correction	15/59	+B: MAX-HI

Except European (L.H.D. U.K.) specs.

Correction	Set temperature	Blower motor applied voltage
MAX HOT correction	32/90	12.1 (V): AUTO-HI
MAX COLD correction	18/64	+B: MAX-HI

Start-up window fogging prevention correction

- The start-up window fogging prevention correction prevents the windows from fogging when all of the following conditions are met.
  - — Right after engine start
  - — Heater is operated
  - — Air is blowing from vents for defrosting

Defroster correction

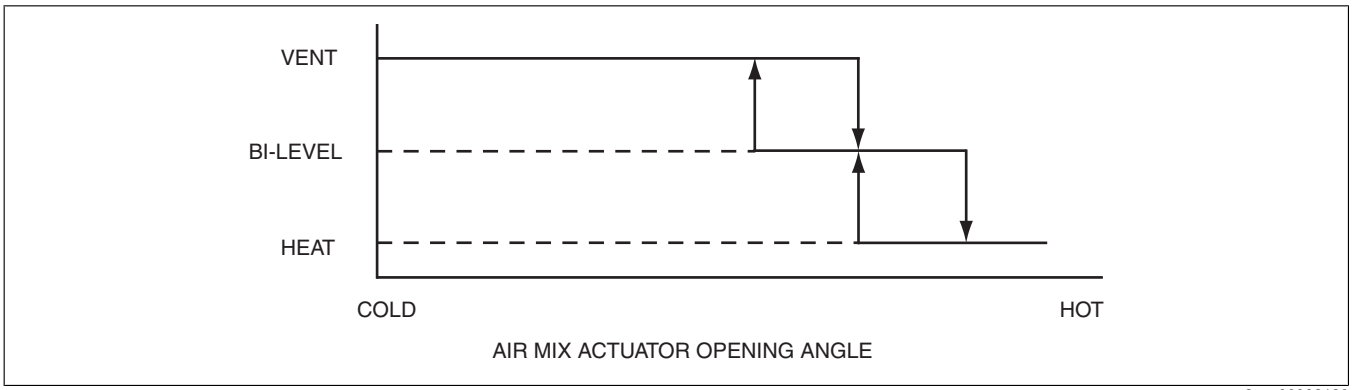
- When the airflow mode is set to the defroster, the defroster correction adds a correction **+2 V** to the blower motor applied voltage to increase airflow volume.

Airflow mode control

- The airflow mode control drives the airflow mode actuator in conjunction with the MODE dial and the vehicle conditions, and switches the position of the airflow mode door.
- The airflow mode control has automatic and manual controls.
- The airflow mode automatic control performs the following correction:
  - — Engine coolant temperature correction (warm-up correction)

Airflow mode automatic control

- The climate control unit determines the airflow mode based on the current air mix actuator opening angle,



am2zzn00002129

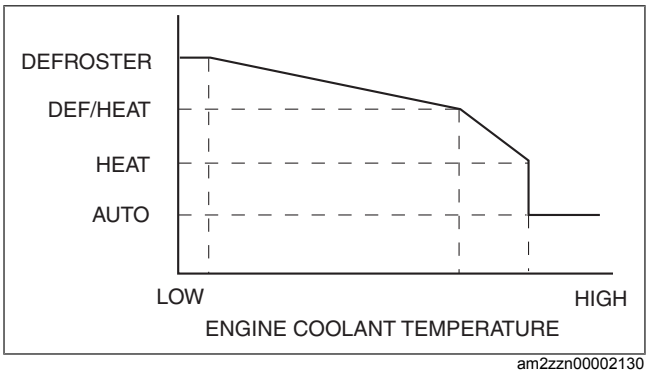
Airflow mode manual control

- The climate control unit switches the airflow mode by the operation of the MODE dial.

Airflow mode	Air vent
VENT	Center vent, side vent
BI-LEVEL	Center vent, side vent, front heat
HEAT	Side vent (low volume), front heat, side demister (low volume), defroster (low volume)
DEF/HEAT	Side vent (low volume), front heat, side demister, defroster
DEFROSTER	Side vent (low volume), side demister, defroster

Engine coolant temperature correction (warm-up correction)

- The engine coolant temperature correction (warm-up correction) prevents cold air from coming out of the lower vents after engine start in winter
- The engine coolant temperature correction (warm-up correction) switches the airflow mode in conjunction with the increase in the engine coolant temperature.



am2zzn00002130

Air intake control

- The air intake control drives the air intake actuator and switches the air intake door position in conjunction with the operation of the MODE dial and the vehicle conditions.
- The air intake control performs the following control:
  - — Air intake manual control
  - — Air intake actuator motor protection function
  - — RECIRCULATE mode control under severe heat
  - — FRESH control in defroster mode

Air intake manual control

- The climate control unit switches the air intake mode by the operation of the RECIRCULATE switch.

Air intake mode (RECIRCULATE switch operation)	RECIRCULATE indicator light
FRESH → RECIRCULATE	On
RECIRCULATE → FRESH	Off

Air intake actuator motor protection function

- For the air intake actuator motor protection function, if the climate control unit detects an excessive load on the motor while the air intake actuator is operating, it stops the air intake actuator motor.

RECIRCULATE mode control under severe heat

- When the vehicle is driven at a very low speed with the A/C on under severe heat conditions, the climate control unit may change the air intake mode from FRESH to RECIRCULATE automatically.
- Due to this, load on the A/C compressor is reduced.
- When the air intake mode is switched to RECIRCULATE, the indicator on the RECIRCULATE switch turns on.

FRESH control in defroster mode

- The FRESH control in the defroster mode switches the intake mode to FRESH when the airflow mode is set to the defroster.

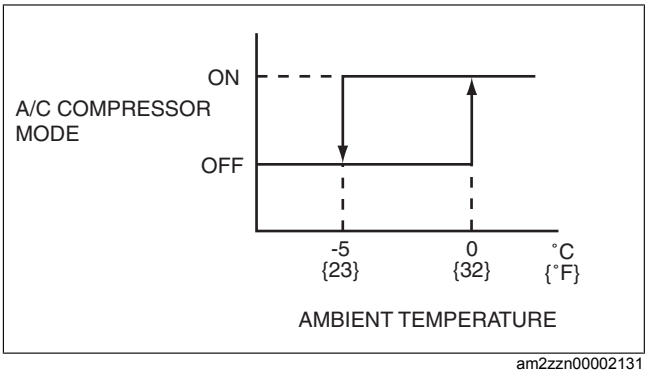
- The FRESH control in the defroster mode improves the effect of defrosting by switching the air intake mode to FRESH.
- The FRESH control in the defroster mode switches the air intake mode to FRESH even if the system was set manually to RECIRCULATE.

A/C compressor control

- The A/C compressor control turns the A/C compressor on/off based on the climate control unit operation and signals from each sensor.
- The A/C compressor control has automatic and manual controls.
- The A/C compressor automatic control performs the following correction:
  — Defroster correction

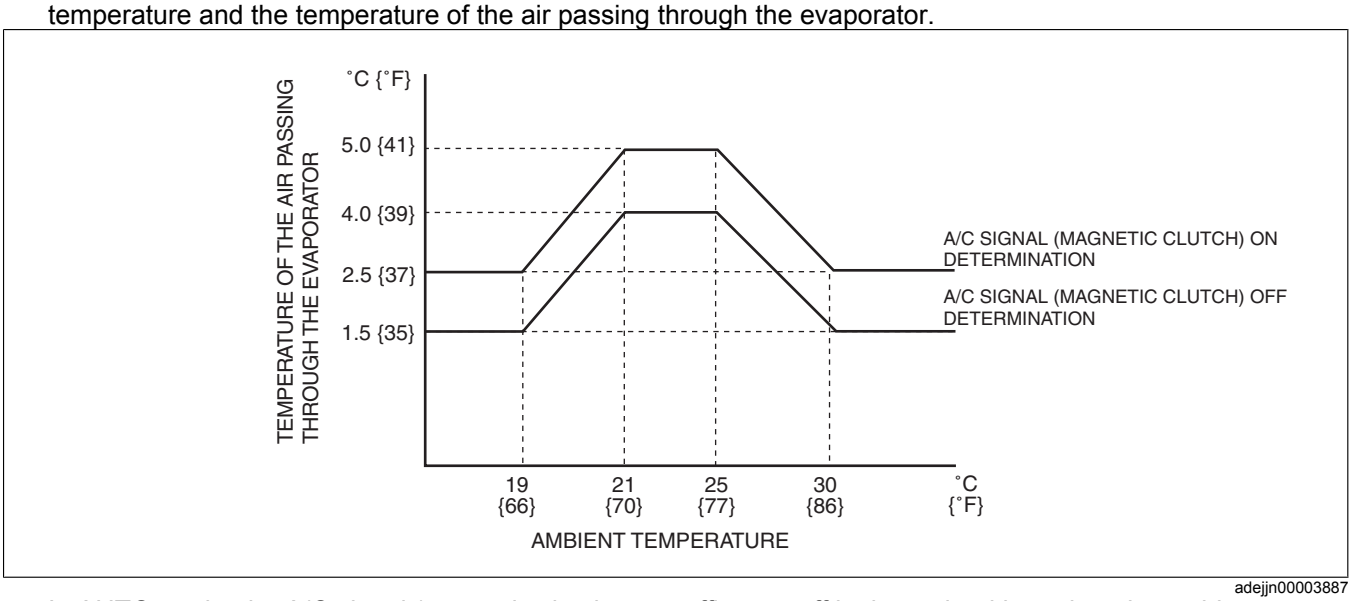
A/C compressor automatic control

- The climate control unit determines the A/C compressor mode on or off based on the ambient temperature.
- The climate control unit determines the A/C signal on or off in conjunction with the air conditioner operation such as A/C on mode (A/C switch on) or AUTO mode (AUTO switch on).



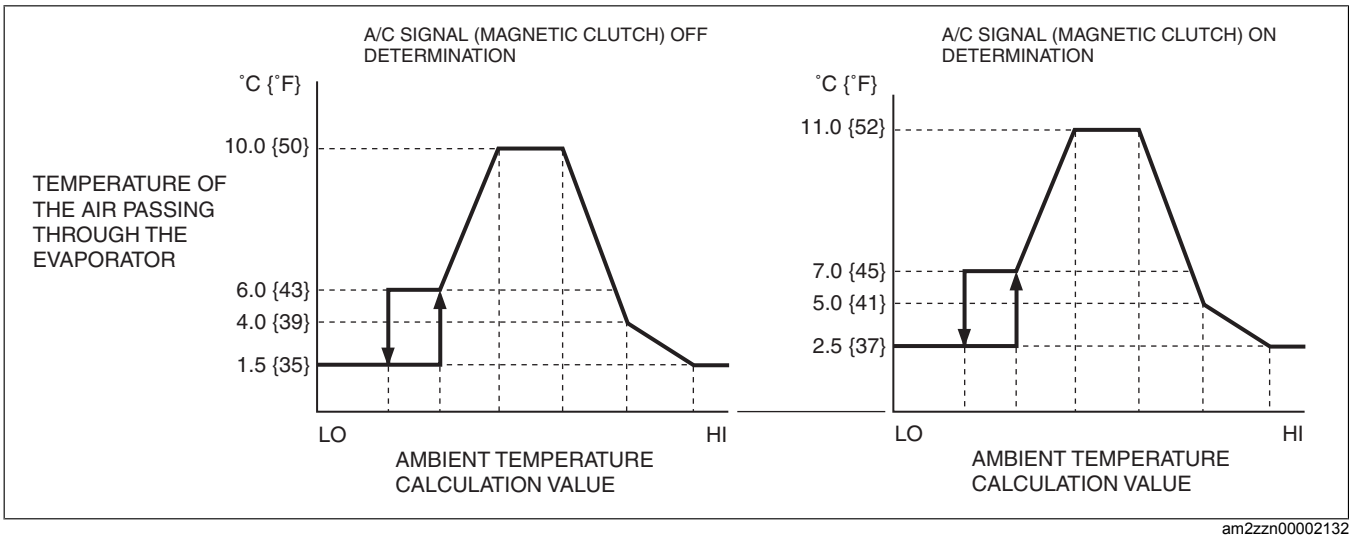
am2zzn00002131

- In the A/C compressor on mode, the A/C signal (magnetic clutch) on/off is determined based on the ambient temperature and the temperature of the air passing through the evaporator.



adejjn00003887

- In AUTO mode, the A/C signal (magnetic clutch on or off) on or off is determined based on the ambient temperature calculation value and the temperature of the air passing through the evaporator.



am2zzn00002132

- The ambient temperature calculation value is calculated based on the ambient temperature, set temperature, and the cabin temperature.
- When the ambient temperature is low, the A/C signal (magnetic clutch) off temperature is set lower to prevent window fogging.

A/C compressor manual control

- The climate control unit switches the A/C compressor mode on or off according to the operation of the A/C switch.

A/C mode (A/C switch operation)	A/C indicator light
OFF→ ON	On
ON → OFF	Off

Defroster correction

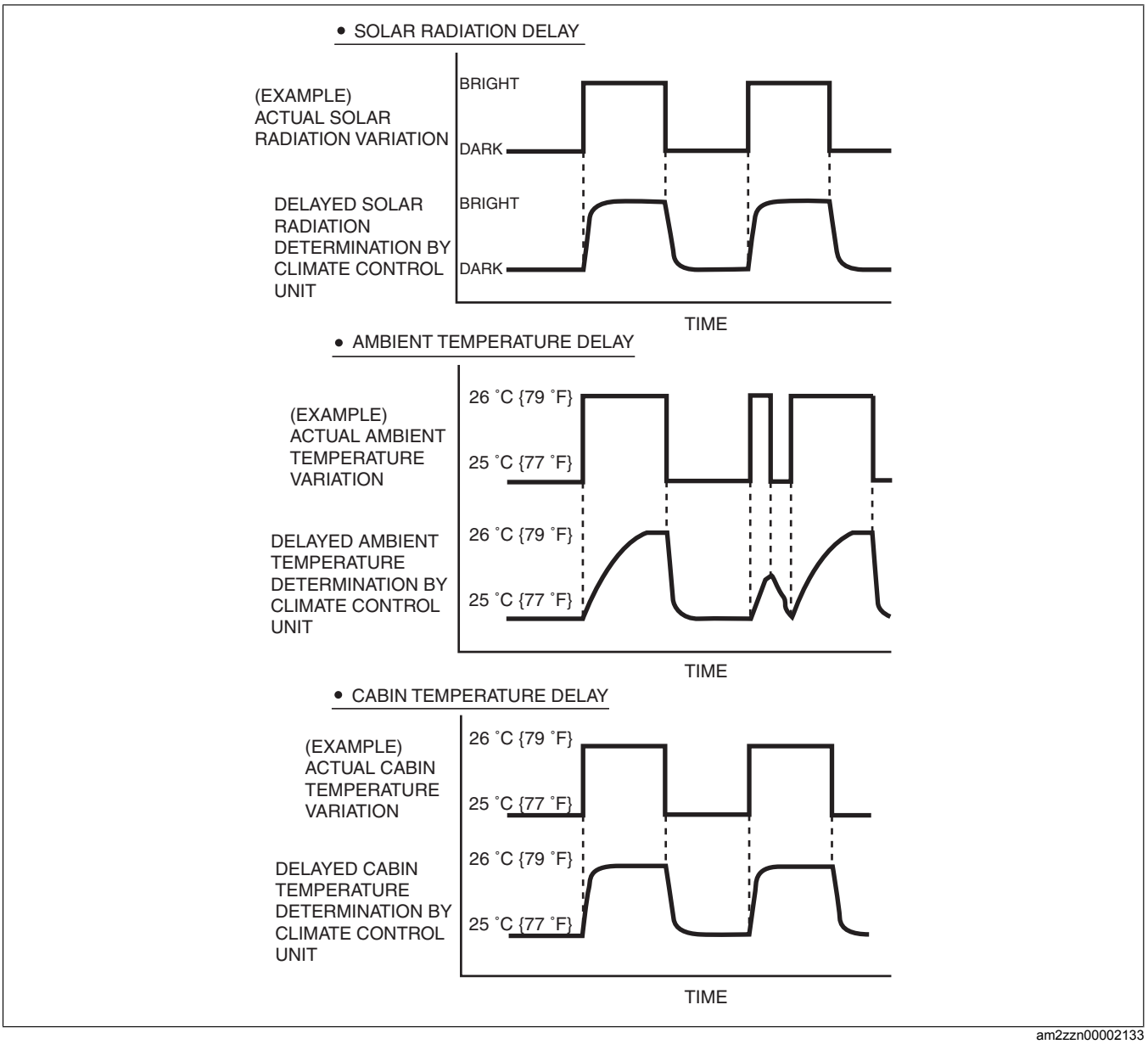
- The defroster correction turns the A/C compressor mode on to prevent window fogging when the airflow mode is set to the defroster.

A/C cut-off control

- The PCM performs A/C cut-off control based on the input signal from the climate control unit and outputs an operation signal to the A/C relay (magnetic clutch). (See A/C CUT-OFF CONTROL [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)(See A/C CUT-OFF CONTROL [SKYACTIV-D 1.5].)

Sensor signal delay function

- The amount of solar radiation, and the ambient and cabin temperatures may change due to factors such as direct and intermittent sunlight (traveling through a city or a highway tunnel), or radiant heat from the ground under parked vehicles as well as the opening and closing of doors.
- If control was performed based exactly on these variations, the air conditioning function would be negatively affected and smooth control could not occur.
- In order to prevent this, the climate control unit delays and smooths the input signals for solar radiation, and the ambient and cabin temperature as shown in the following figure, realizing stable control.

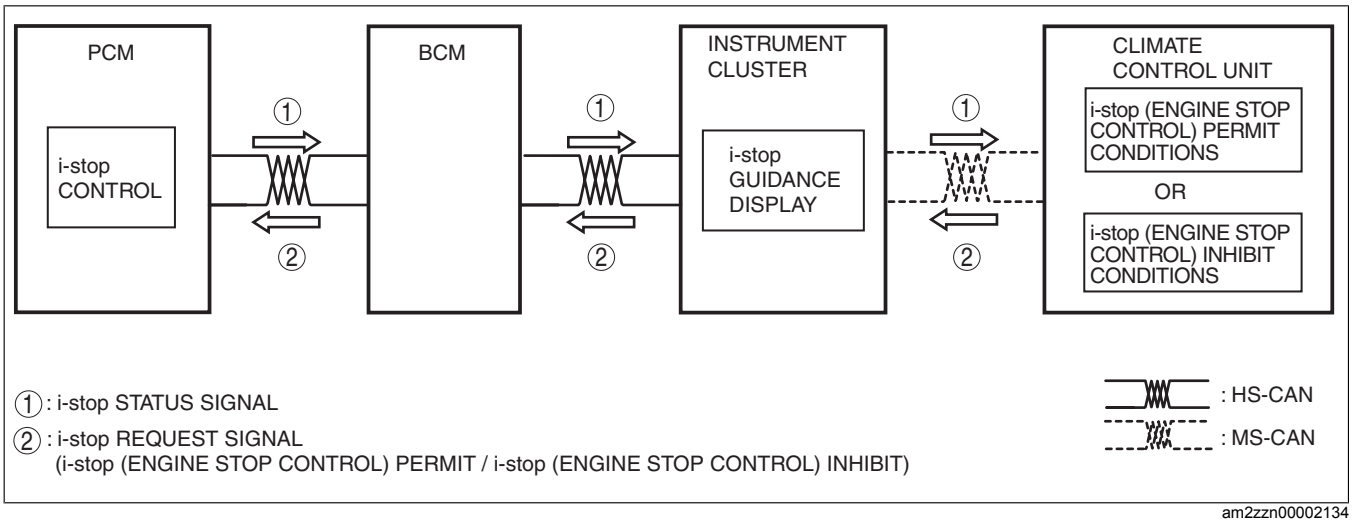


am2zzn00002133

- When the engine is restarted after being temporarily stopped, the ambient temperature sensor may detect a temperature higher than the actual ambient temperature.
- To prevent this, if the engine coolant temperature is **55 °C {131 °F} or more**, each control is performed based on the data for the ambient temperature before the engine was stopped that was stored in the climate control unit.

Air conditioner i-stop control (With i-stop)

- The climate control unit determines i-stop control (engine stop control) according to the operation condition of the air conditioner system and sends i-stop (engine stop control) request signals to the PCM.



am2zzn00002134

- The climate control unit controls the air mix actuator according to the change in the airflow temperature passing through the heater core or the evaporator during i-stop operation (engine stop control) as follows and corrects the airflow temperature.
  - — If airflow temperature passing through heater core decreases, air mix actuator is opened in HOT direction
  - — If airflow temperature passing through evaporator increases, air mix actuator is opened in COLD direction

Caution

- **Occupants may hear the sound of the air mix actuator operating because there is no engine sound during the i-stop operation (engine stop control). The airflow temperature control is performed even during i-stop operation (engine stop control), therefore, it does not indicate a system malfunction even if the operation sound of the air mix actuator is heard.**

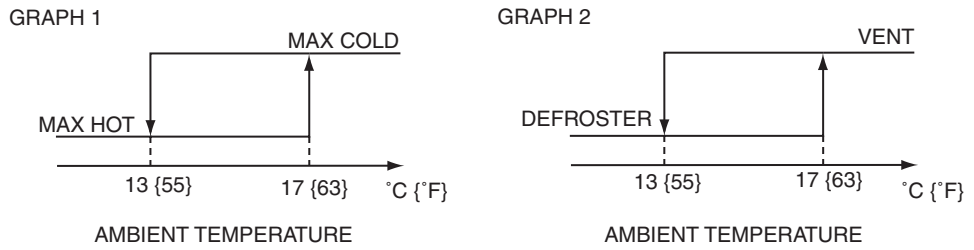
Fail-safe

- If a malfunction is detected by the malfunction detection function, the fail-safe function performs the following controls to prevent an operation malfunction of the auto air conditioner and malfunction of output devices.

Fail-safe function table

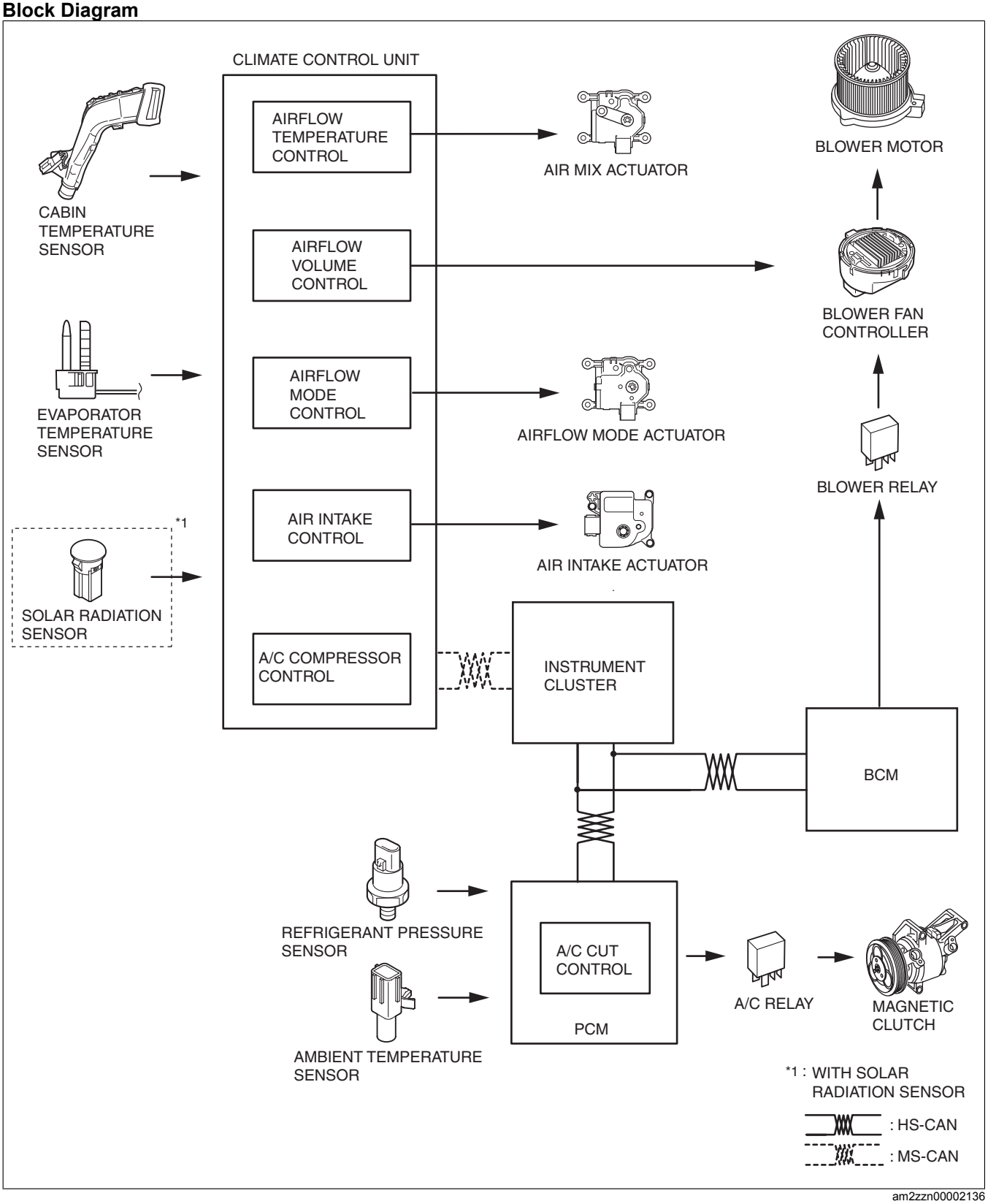
Part where malfunction is determined	Malfunction determined when ignition is switched ON	Ignition switched ON under malfunction condition
Ambient temperature sensor	Ambient temperature sensor input value is fixed at value right before malfunction	Ambient temperature sensor input value is fixed at **15 °C {59 °F}.**
Cabin temperature sensor	Cabin temperature sensor input value is fixed at the value right before the malfunction.	Cabin temperature sensor input value is fixed at **25 °C {77 °F}.**
Evaporator temperature sensor	Evaporator temperature sensor input value fixed at **0 °C {32 °F}**	←
Solar radiation sensor	Solar radiation sensor input value fixed at value right before the malfunction (Solar radiation sensor input value **0.25 V** or less or **4.9 V** or more)	Solar radiation sensor set value fixed at **0 W/m^2**
Engine coolant temperature sensor	Engine coolant temperature sensor input value fixed at **85 °C {185 °F}**.	←
Air mix actuator (Potentiometer)	Air mix actuator drive signal is stopped right after the malfunction is determined. • However, fixed at MAX COLD when manually set temperature is at **15/59** and fixed at MAX HOT when manually set temperature is at **29/84** (European (L.H.D. U.K.) specs.) • However, fixed at MAX COLD when manually set temperature is at **18/64** and fixed at MAX HOT when manually set temperature is at **32/90** (except European (L.H.D. U.K.) specs.)	Control based on ambient temperature (Reference: Graph 1) • However, fixed at MAX COLD when manually set temperature is at **15/59** and fixed at MAX HOT when manually set temperature is at **29/84** (European (L.H.D. U.K.) specs.) • However, fixed at MAX COLD when manually set temperature is at **18/64** and fixed at MAX HOT when manually set temperature is at **32/90** (except European (L.H.D. U.K.) specs.)
Airflow mode actuator (Potentiometer)	Airflow mode actuator drive signal is stopped right when the malfunction is determined. • For manual operation, only vent or defroster mode is operable	Control based on ambient temperature (Reference: Graph 2) • For manual operation, only vent or defroster mode is operable

Part where malfunction is determined	Malfunction determined when ignition is switched ON	Ignition switched ON under malfunction condition
Air mix actuator (Motor lock)	• Air mix actuator drive signal is stopped right after the malfunction is determined.	• **20 s** after the ignition is switched ON (engine off or on), the normal air mix actuator drive signal is output again. After that, motor output is stopped if a malfunction is detected.
Airflow mode actuator (Motor lock)	• Airflow mode actuator drive signal is stopped right when the malfunction is determined.	• **9 s** after the ignition is switched ON (engine off or on), the normal mode actuator drive signal is output again. After that, motor output is stopped if a malfunction is detected.



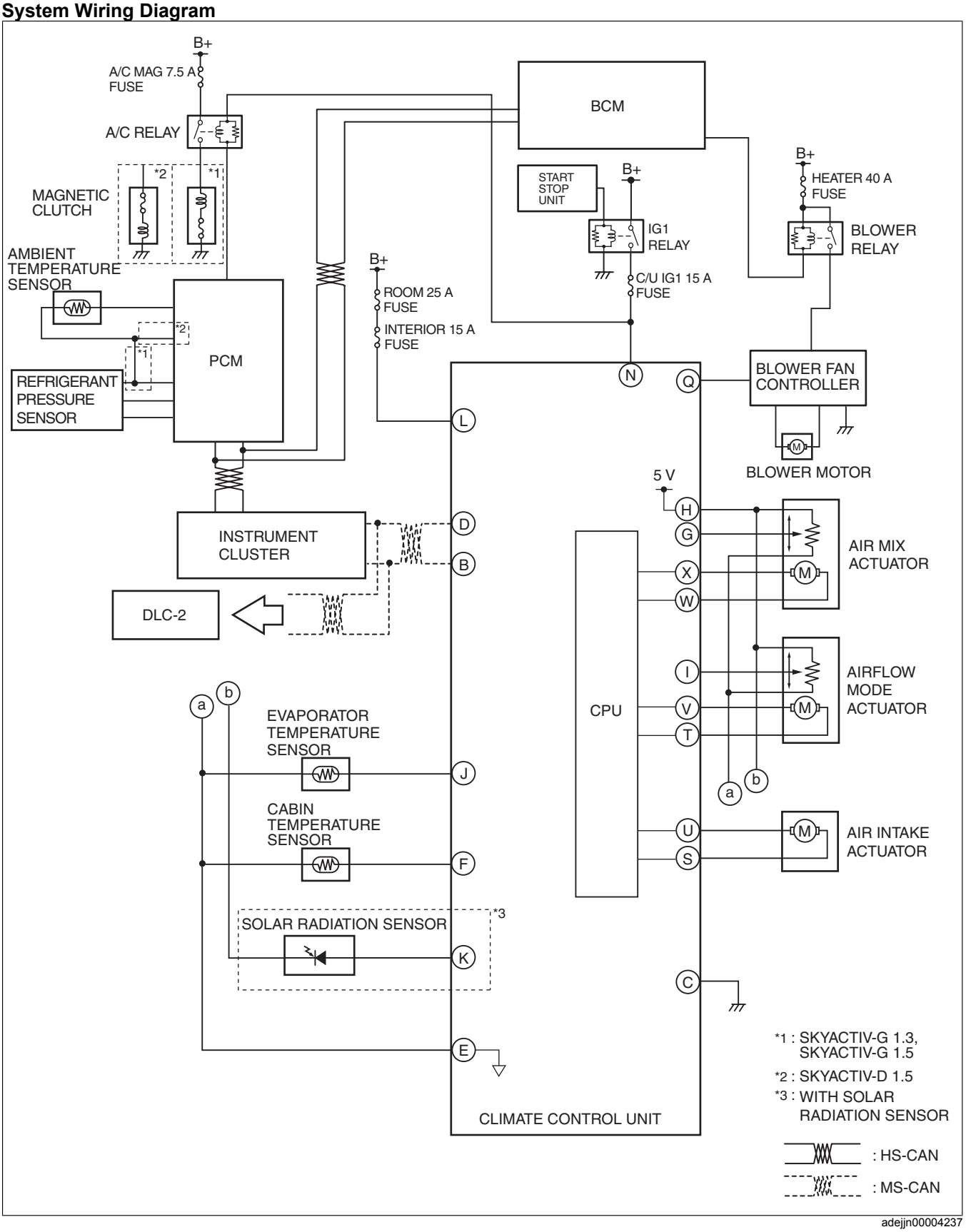
am2zzn00002135

Block Diagram



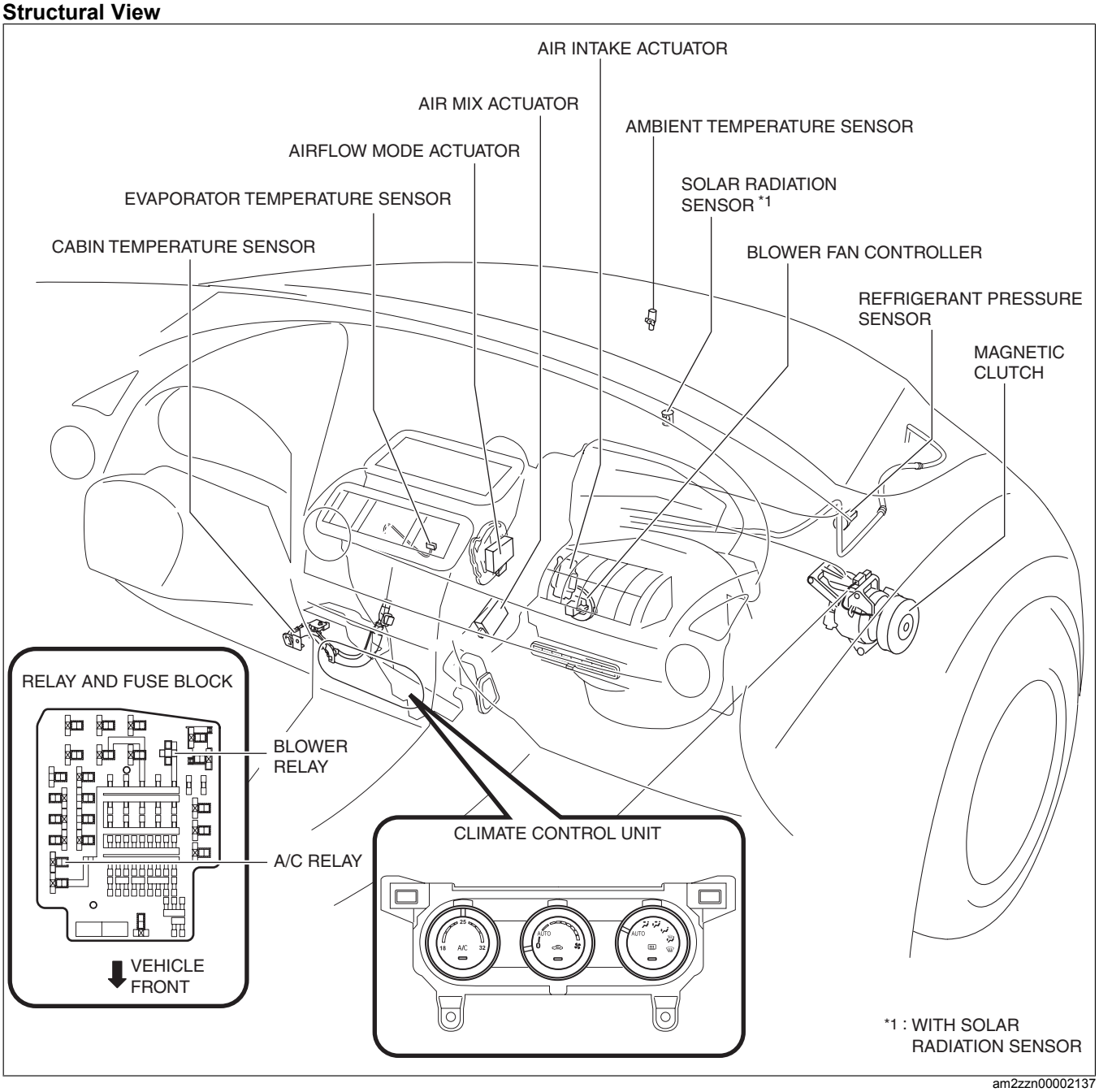
am2zzn00002136

System Wiring Diagram



adejjn00004237

Structural View

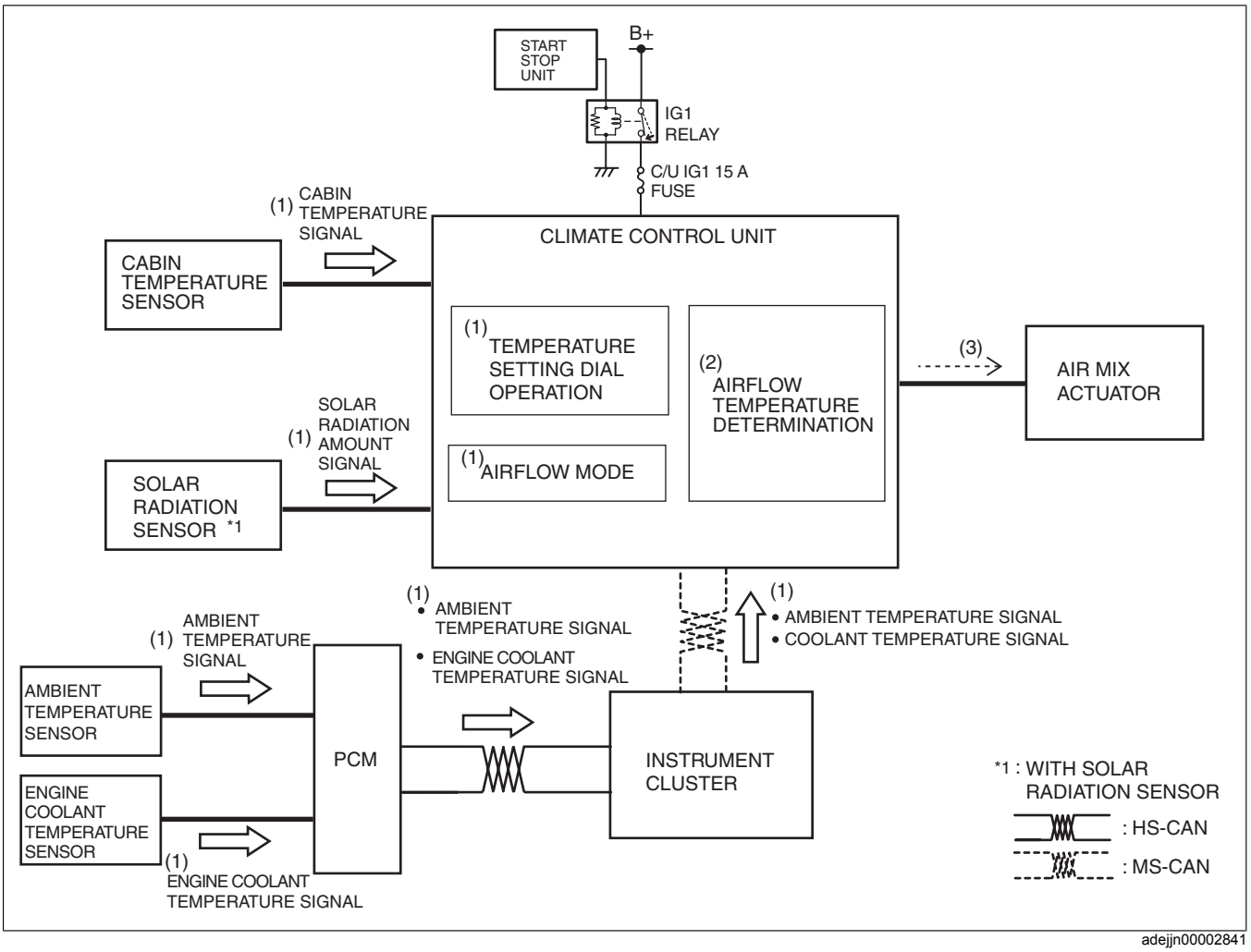


am2zzn00002137

Operation

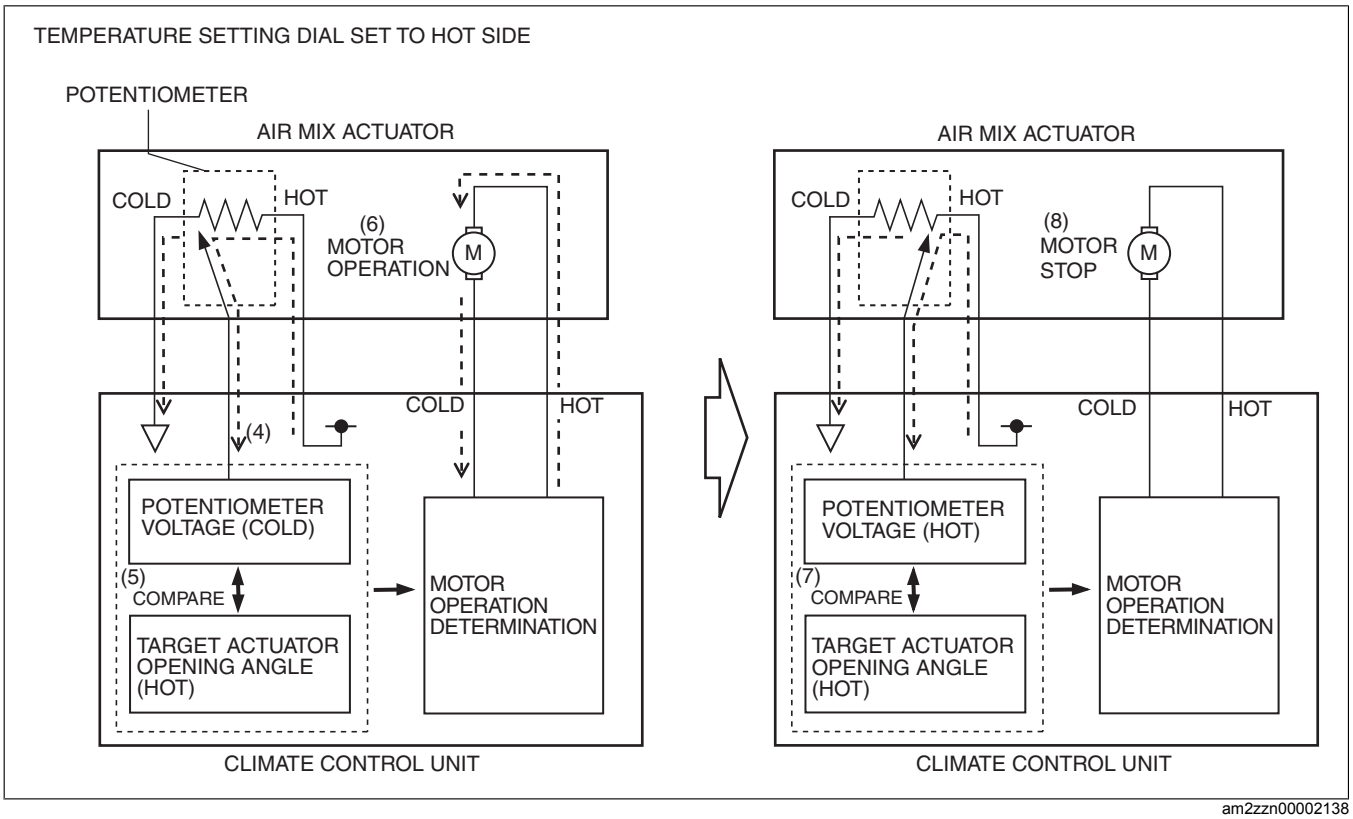
Airflow temperature control

1. The climate control unit performs airflow temperature determination (2) based on the signals (1) from each sensor which change according to the temperature setting dial operation and the vehicle conditions when the ignition is switched ON (engine off or on).
2. The climate control unit drives (3) the air mix actuator based on the results of the airflow temperature determination and corrections.



adejjn00002841

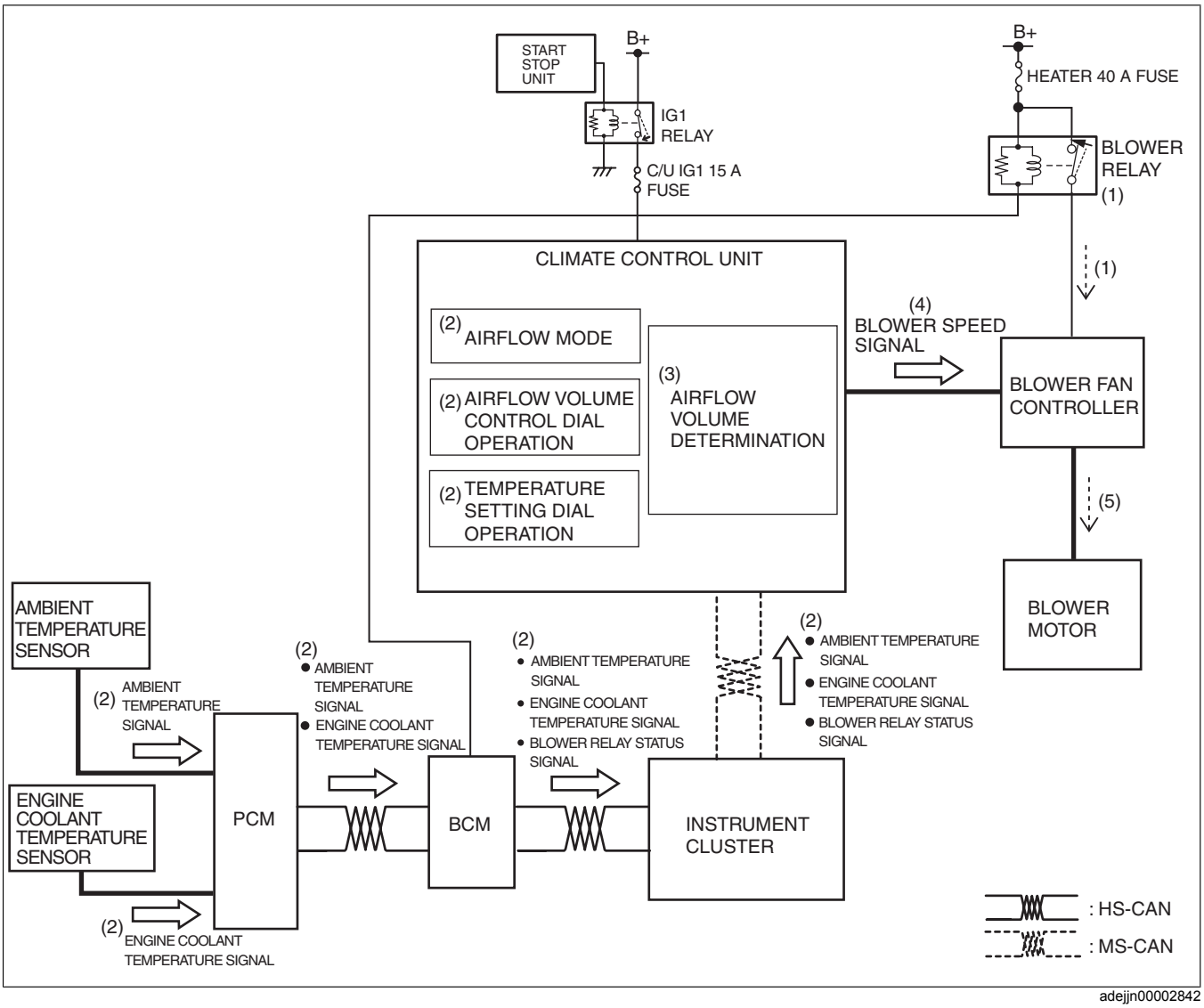
3. The climate control unit detects the current air mix actuator opening angle based on the potentiometer voltage (4).
4. The climate control unit compares the target actuator opening angle with the voltage from the potentiometer.
5. When the potentiometer voltage value differs from the target actuator opening angle, the climate control unit drives (6) the air mix actuator motor.
6. When the potentiometer voltage value matches (7) the target actuator opening angle, the climate control unit stops (8) the air mix actuator motor.



am2zzn00002138

Airflow volume control

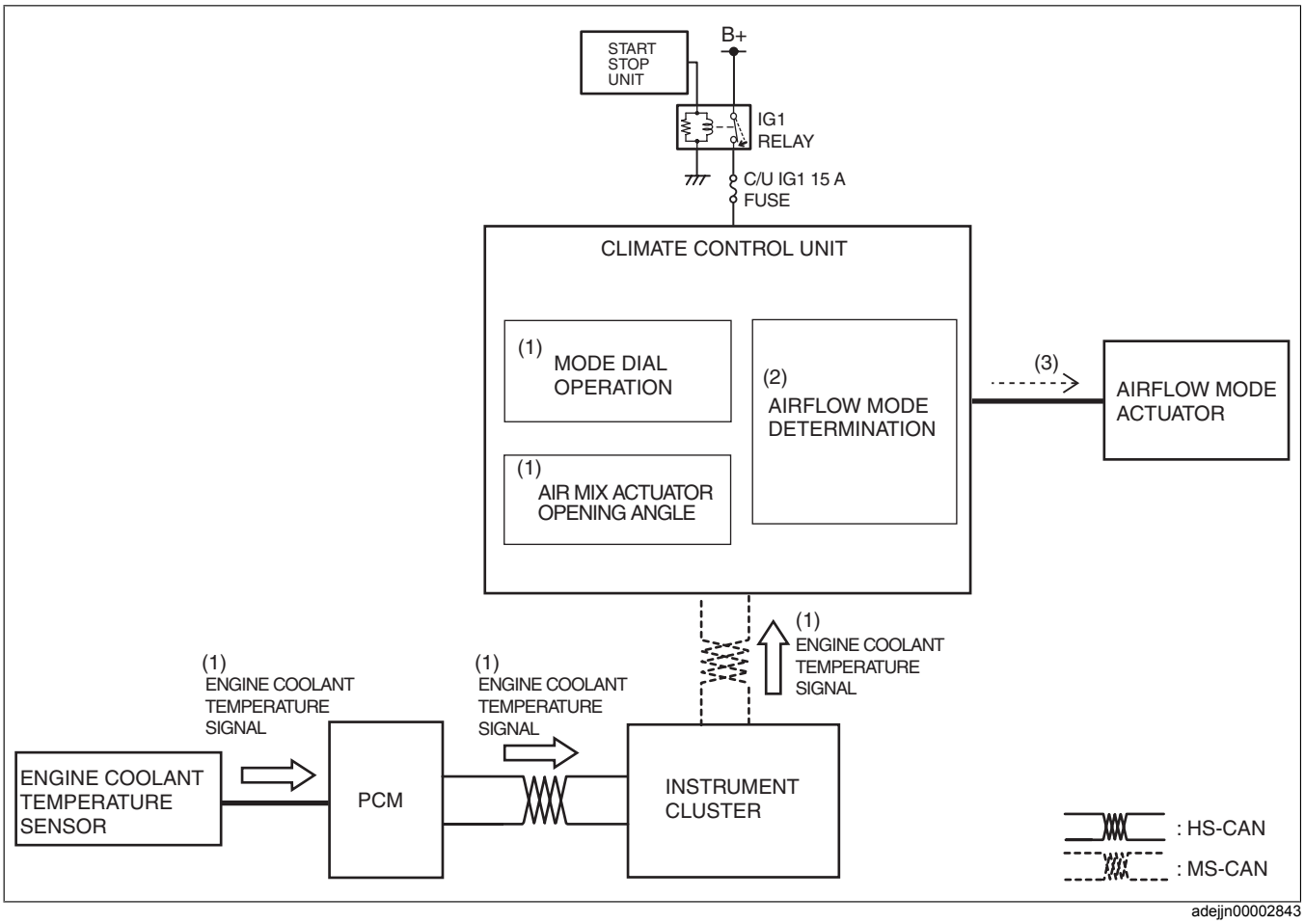
1. When the ignition is switched ON (engine off or on), the BCM turns the blower relay on (1).
2. When the blower relay turns on, the blower fan controller goes on operation standby.
3. The climate control unit performs airflow volume determination (3) based on the operation of the airflow volume control dial and the temperature setting dial, the airflow mode condition, and the signals from each sensor when the ignition is switched ON (engine off or on).
4. The climate control unit sends a blower speed signal (4) to the blower fan controller based on the results of the airflow volume determination and the corrections.
5. The blower fan controller operates (5) the blower motor based on the blower speed signal.



adejjn00002842

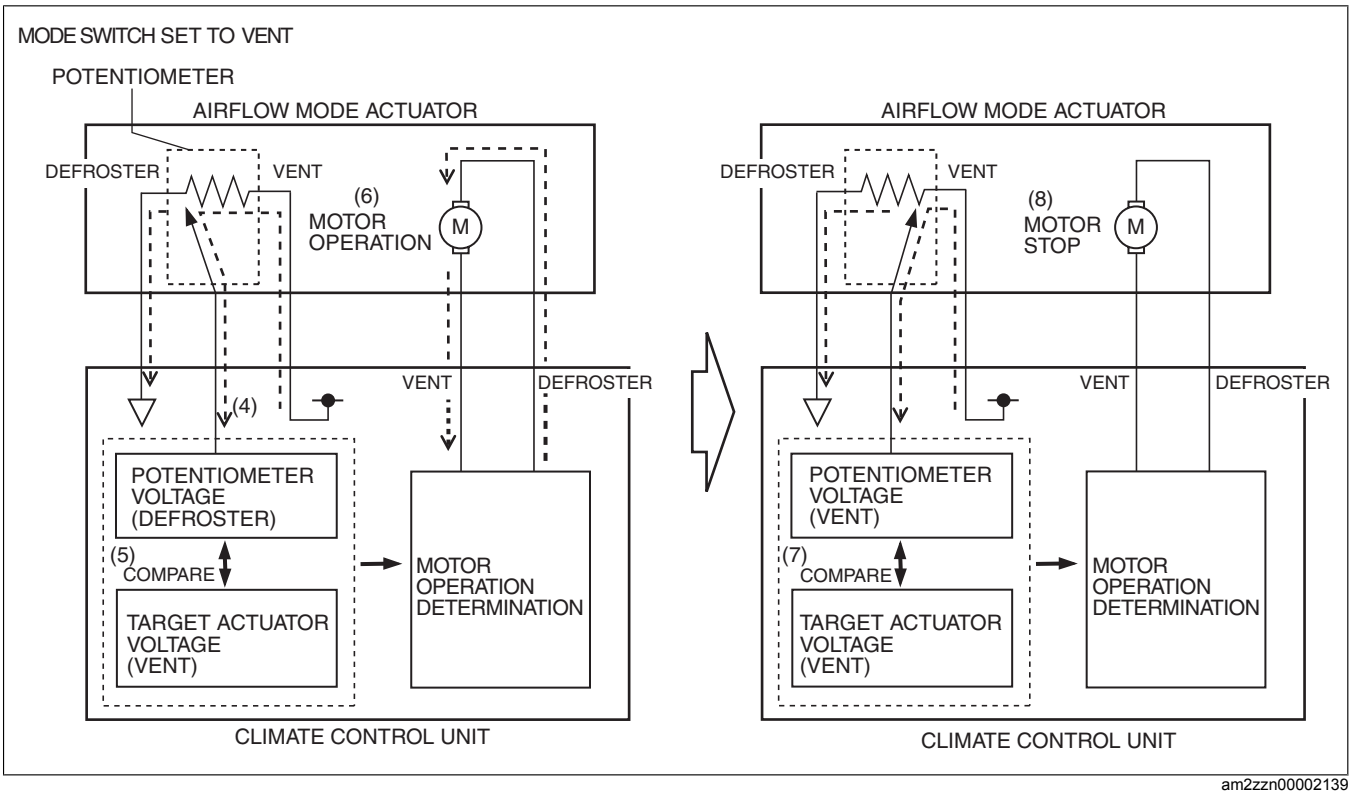
Airflow mode control

1. The climate control unit performs airflow mode determination (2) based on the signals (1) from each sensor which change according to the MODE dial operation and the vehicle conditions when the ignition is switched ON (engine off or on).
2. The climate control unit drives (3) the airflow mode actuator based on the results of the airflow mode determination and corrections.



adejjn00002843

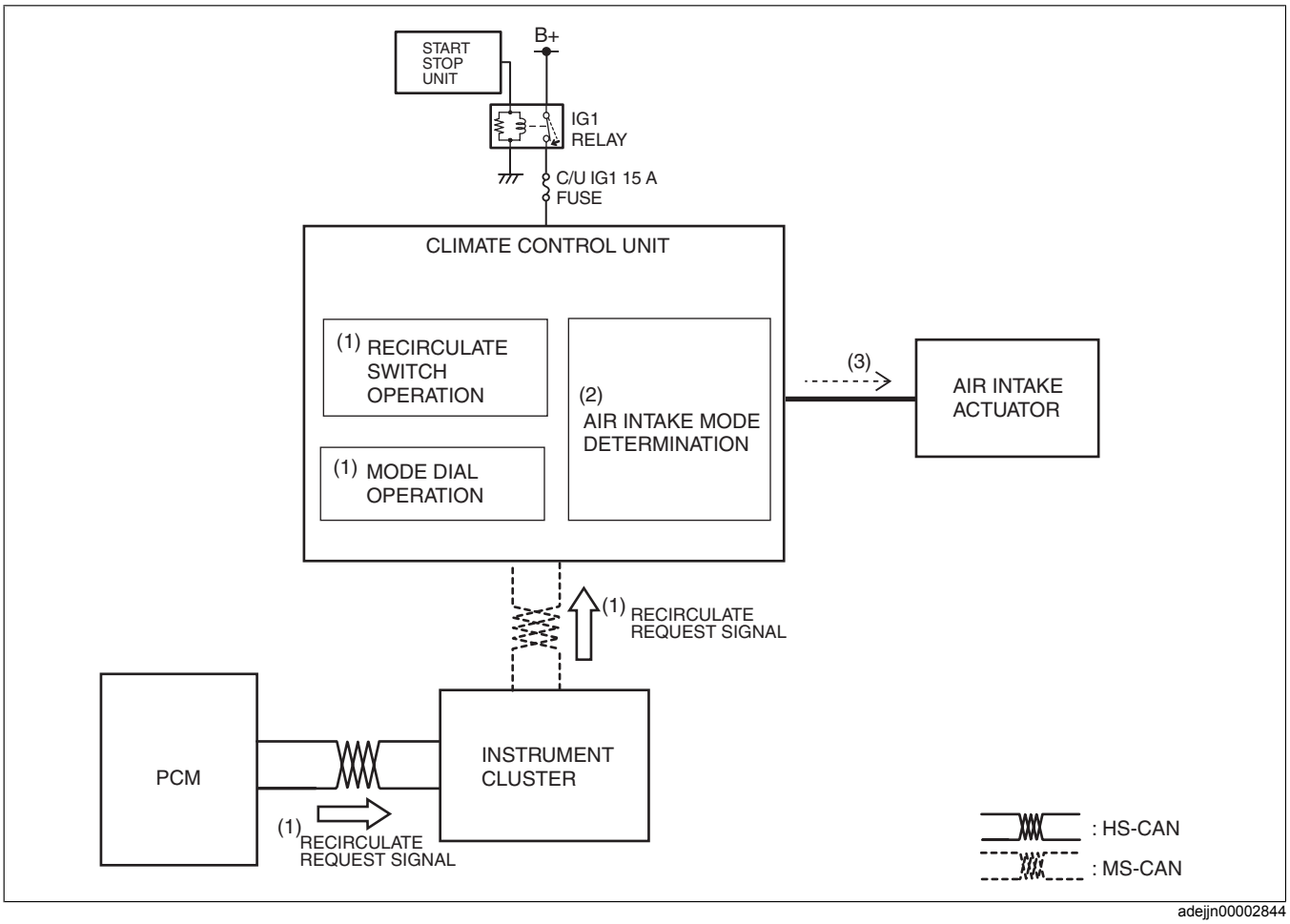
3. The climate control unit detects the current airflow mode actuator opening angle based on the potentiometer voltage (4).
4. The climate control unit compares the target actuator voltage value with the voltage from the potentiometer.
5. When the potentiometer voltage value differs from the target actuator voltage value, the climate control unit drives (6) the airflow mode actuator motor.
6. When the potentiometer voltage value matches (7) the target actuator voltage value, the climate control unit stops (8) the airflow mode actuator motor.



am2zzn00002139

Air intake control

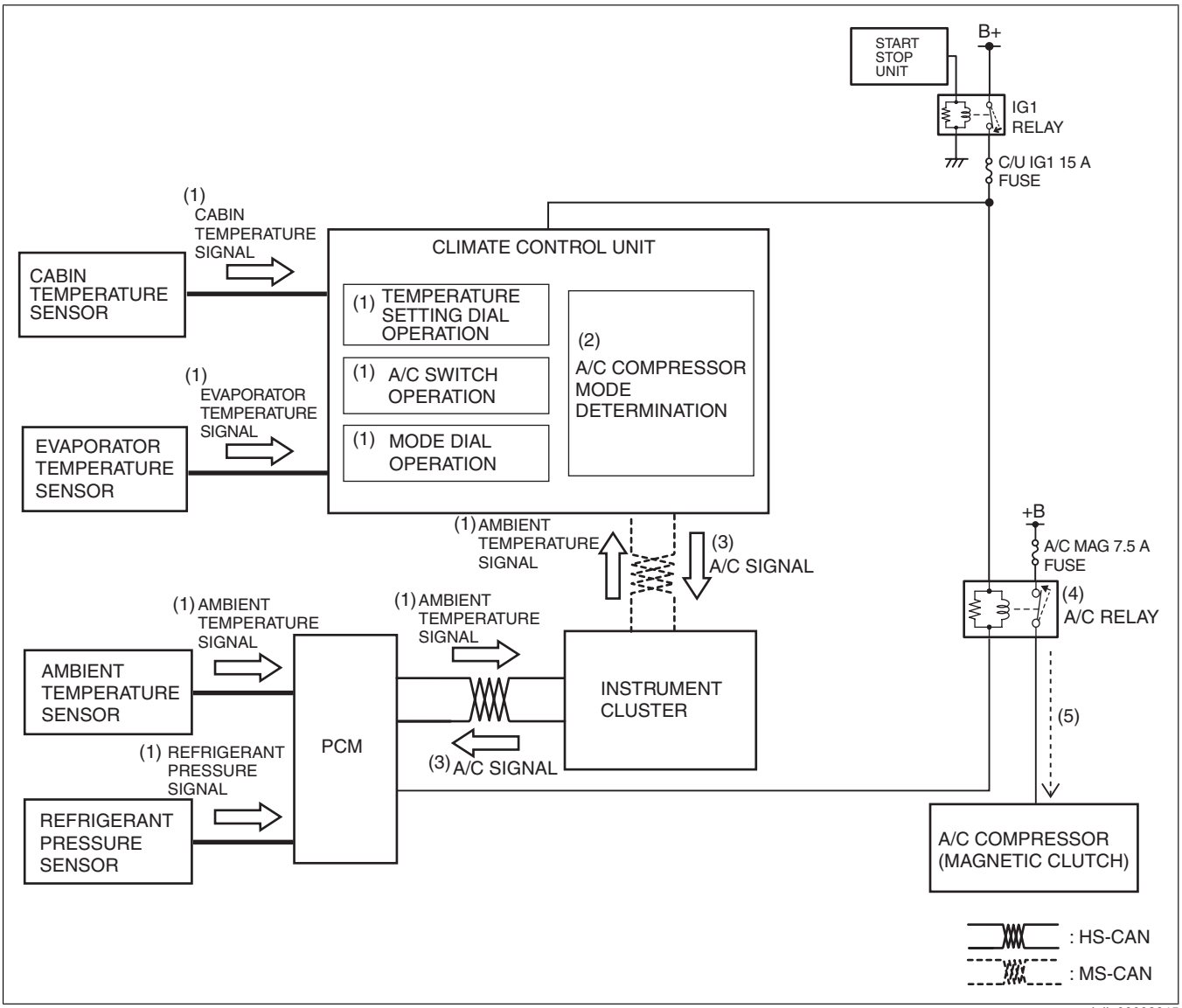
1. The climate control unit performs air intake mode determination (2) based on the signals (1) from the operation of the RECIRCULATE switch and the MODE dial and the PCM when the ignition is switched ON (engine off or on).
2. The climate control unit drives (3) the air intake actuator for a specified time based on the results of the air intake mode determination and corrections.



adejjn00002844

A/C compressor control

1. The climate control unit performs A/C compressor mode determination (2) based on the signals (1) from each sensor which change according to each switch/dial operation and the vehicle conditions when the ignition is switched ON (engine off or on).
2. The climate control unit sends (3) the A/C signal to the PCM according to the result of the A/C compressor mode determination and corrections.
3. The PCM turns the A/C relay on (4) based on the signals from each sensor which change according to the A/C signal and the vehicle conditions.
4. When the A/C relay turns on, the magnetic clutch turns on (5).



adejjn00002845

Air conditioner i-stop control (With i-stop)

i-stop (engine stop control) request control (i-stop inhibit/i-stop permit)

- The climate control unit sends an i-stop inhibit or permit request signal to the PCM according to the operation conditions of the air conditioner system.
- If any of the following i-stop inhibit conditions is met, the climate control unit sends an i-stop inhibit request signal to the PCM.

i-stop inhibit conditions

1	Climate control unit detects DTC		
2	Ambient temperature is **-10 °C {14 °F} or less**, or **50 °C {122 °F} or more**		
3	Blower motor is operating	Defroster is being operated manually	
4		A/C switch is ON	Set temperature is MAX COLD
5			Airflow mode is vent or bi-level
6		Large difference between set temperature and cabin temperature and difference between target airflow temperature and current airflow temperature	

- When all of the i-stop inhibit conditions are met, the climate control unit sends an i-stop permit request signal to the PCM.

Air conditioner system control during i-stop (engine-stop control)

- When the climate control unit determines an engine-stop condition based on the i-stop condition signal sent from the PCM, it starts the air conditioner system control when the engine is stopped.
- When the engine is stopped by the i-stop control, the climate control unit stops the air intake actuator at the current position.

Note

- If the engine is stopped by the i-stop control with the A/C switch turned on, the indicator on the A/C switch is kept turned on.

- Based on the following signals, the climate control unit calculates the engine coolant temperature correction value for the engine-stop condition by i-stop control.
  - Engine coolant temperature signal at moment engine is stopped which is sent from PCM via CAN communication
  - Heater core temperature

Recovery to normal air conditioner system control

- If any of the following i-stop inhibit conditions is met during air conditioner system control when the engine is stopped, the climate control unit sends an engine re-start request signal to the PCM.
- After sending an engine restart request signal, the climate control unit returns to normal air conditioner system control when the engine is restarted.